SIMATS ENGINEERING
ASSESSMENT TOOL 3 - LOW (Coding Challenge)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3, BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks: 50**

ANNEXURE A

CODING CHALLENGE

Code of Conduct:

I S.K.Deepa Arathi (192571020) that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

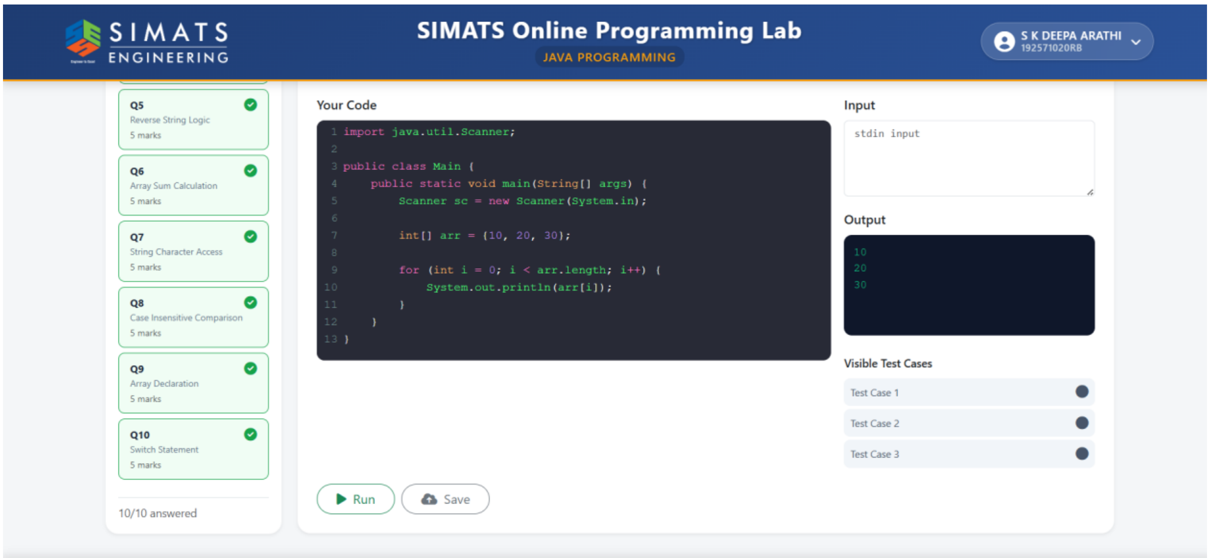
Signature of the student:

Rubrics for Evaluation:

Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Total (30%)	Total (100)
Weightage	20%	25%	25%	20%	15%		
Q1- Q2							
Q3-Q4							
Q5-Q6							
Q7-Q8							
Q9-Q10							
Total							

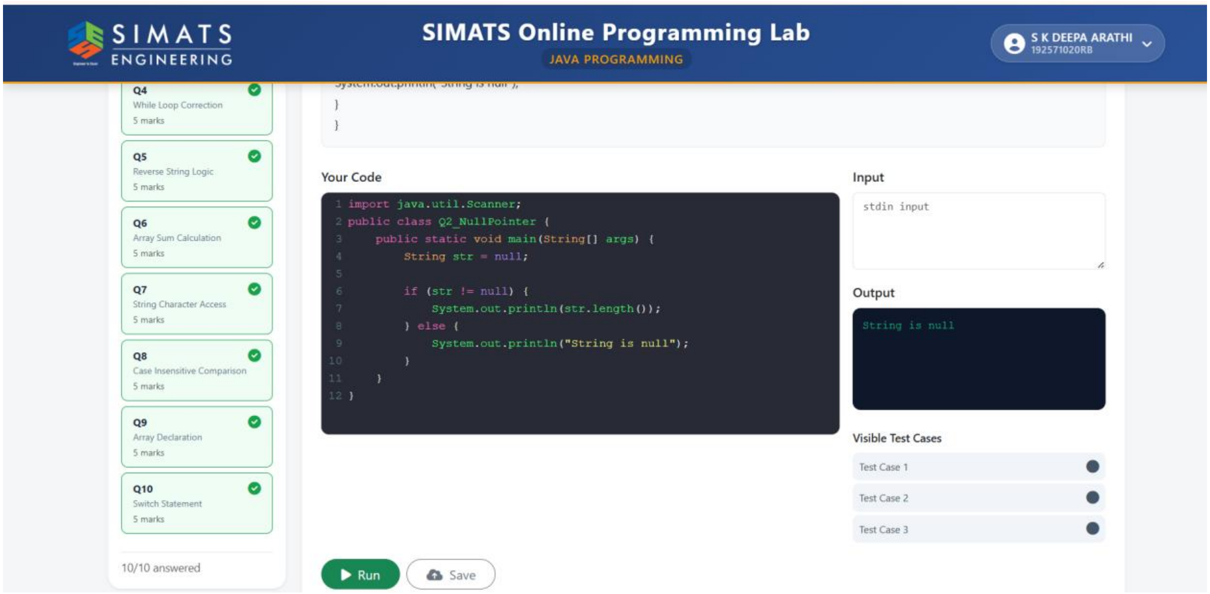
Q1. The following Java program is intended to print all elements of an array. However, it throws an error during execution. Identify and correct the mistake in the loop condition.

```
public class Q1_ArrayIndex {
   public static void main(String[] args) {
      int arr[] = {10, 20, 30};
      for(int i = 0; i < arr.length; i++){
         System.out.println(arr[i]);
      }
   }
}
```



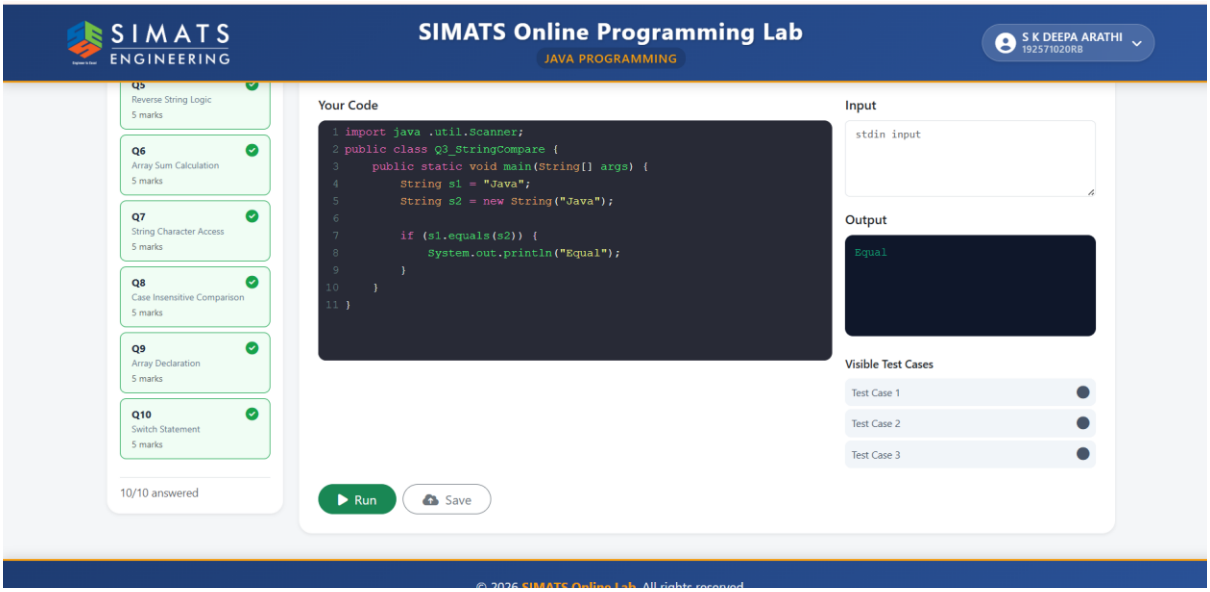
Q2. A program attempts to print the length of a string but results in a runtime exception. Analyze the issue and modify the code to handle it properly.

```
public class Q2_NullPointer {
   public static void main(String[] args) {
      String str = null;
      if(str != null)
         System.out.println(str.length());
      else
         System.out.println("String is null");
   }
}
```



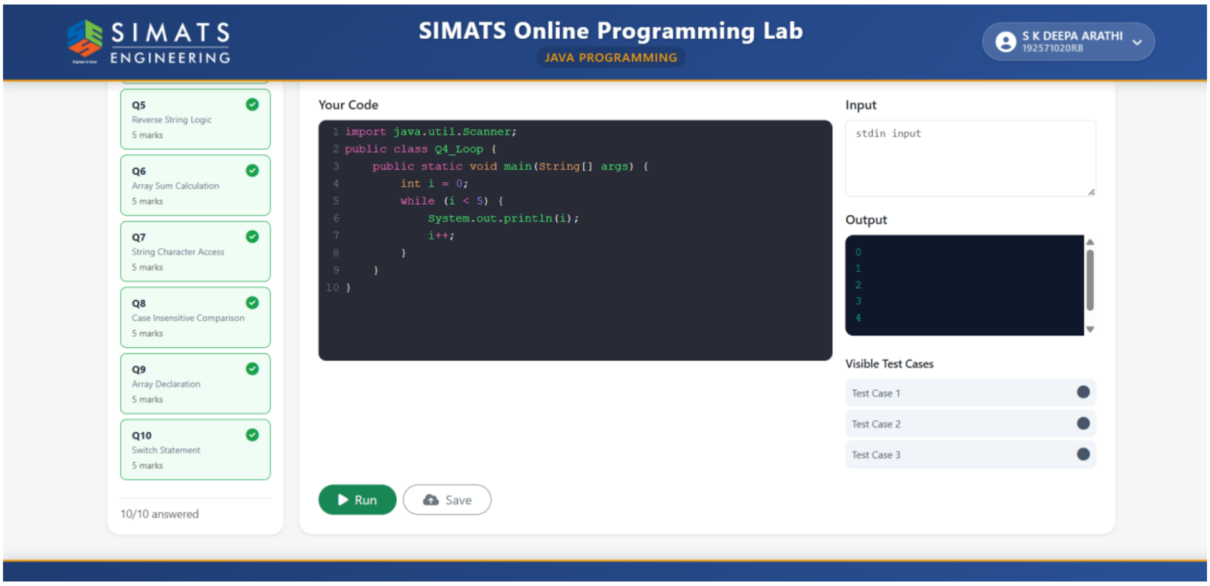
Q3. A developer compares two strings using the == operator, which leads to incorrect output. Identify the issue and correct the comparison method.

```
public class Q3_StringCompare {
   public static void main(String[] args) {
      String s1 = "Java";
      String s2 = new String("Java");

      if(s1.equals(s2)){
         System.out.println("Equal");
      }
   }
}
```



Q4. The given program uses a loop to print numbers from 0 to 4, but it runs infinitely. Identify the cause and fix the loop.

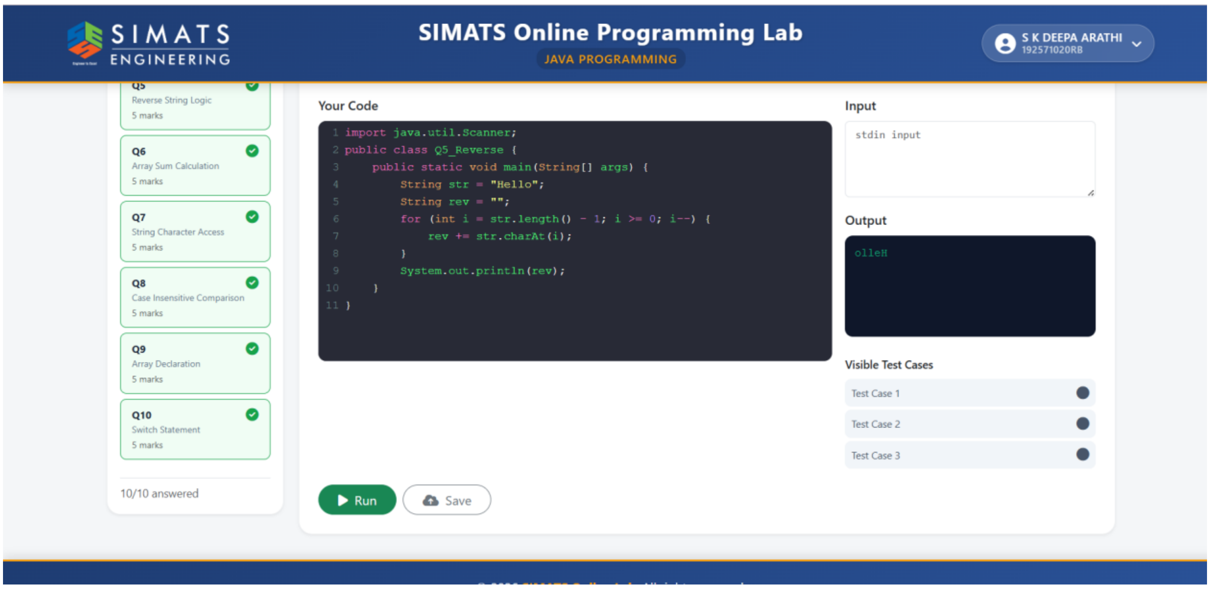
```
public class Q4_Loop {
    public static void main(String[] args) {
        int i = 0;
        while(i < 5){
            System.out.println(i);
            i++;
        }
    }
}
```



Q5. A program is written to reverse a string. Verify whether the logic is correct and modify it if necessary.

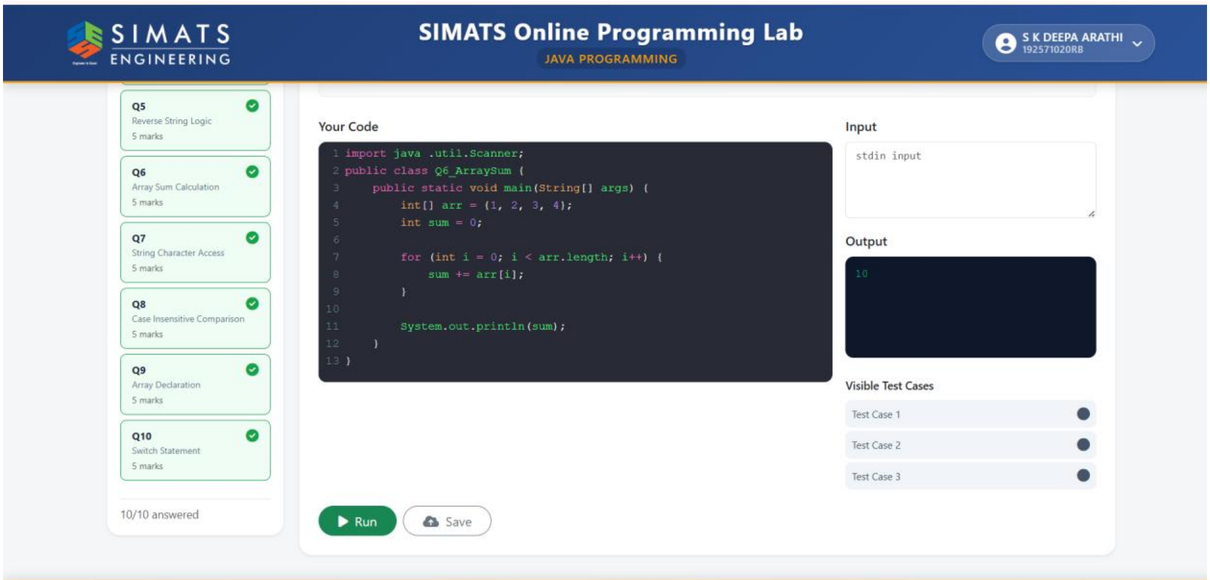
```
public class Q5_Reverse {
    public static void main(String[] args) {
        String str = "Hello";
        String rev = "";

        for(int i = 0; i < str.length(); i++){
            rev = str.charAt(i) + rev;
        }
        System.out.println(rev);
    }
```

}



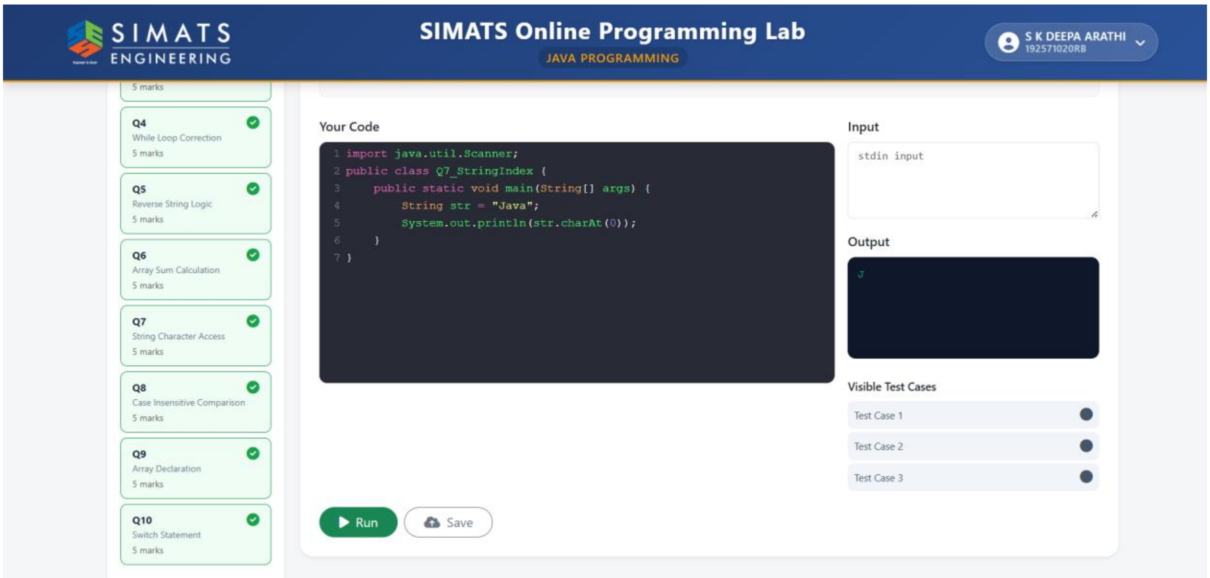
Q6. Identify the problem and fix it.

```
public class Q6_ArraySum {
    public static void main(String[] args) {
        int arr[] = {1,2,3,4};
        int sum = 0;
        for(int i=0;i<arr.length;i++){
            sum += arr[i];
        }
        System.out.println(sum);
    }
}
```



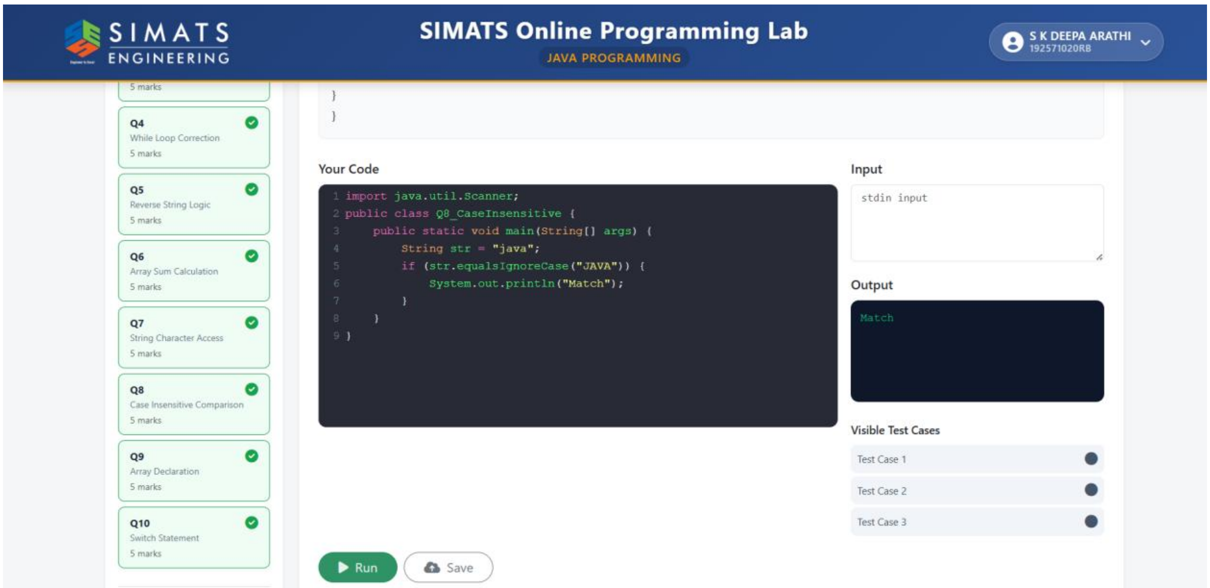
Q7. A program tries to access a character from a string using an invalid index, causing an exception. Identify and correct the issue.

```
public class Q7_StringIndex {
    public static void main(String[] args) {
        String str = "Java";
        System.out.println(str.charAt(3));
    }
}
```



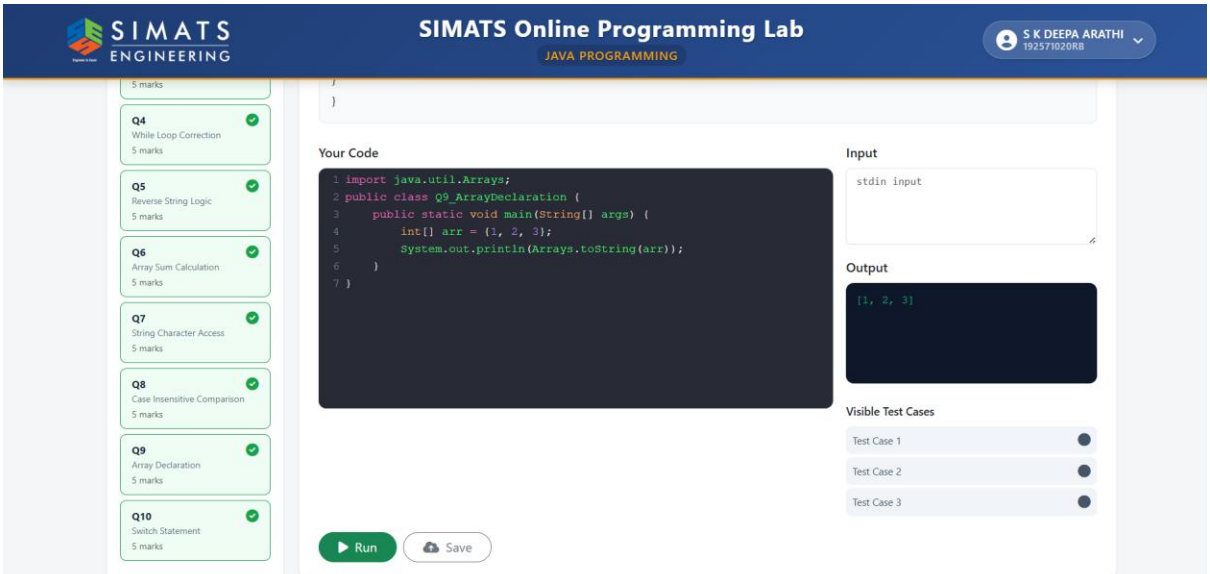
Q8. A string comparison fails due to case sensitivity. Modify the program to make the comparison case-insensitive.

```
public class Q8_CaseInsensitive {
    public static void main(String[] args) {
        String str = "java";
        if(str.equalsIgnoreCase("JAVA")){
            System.out.println("Match");
        }
    }
}
```



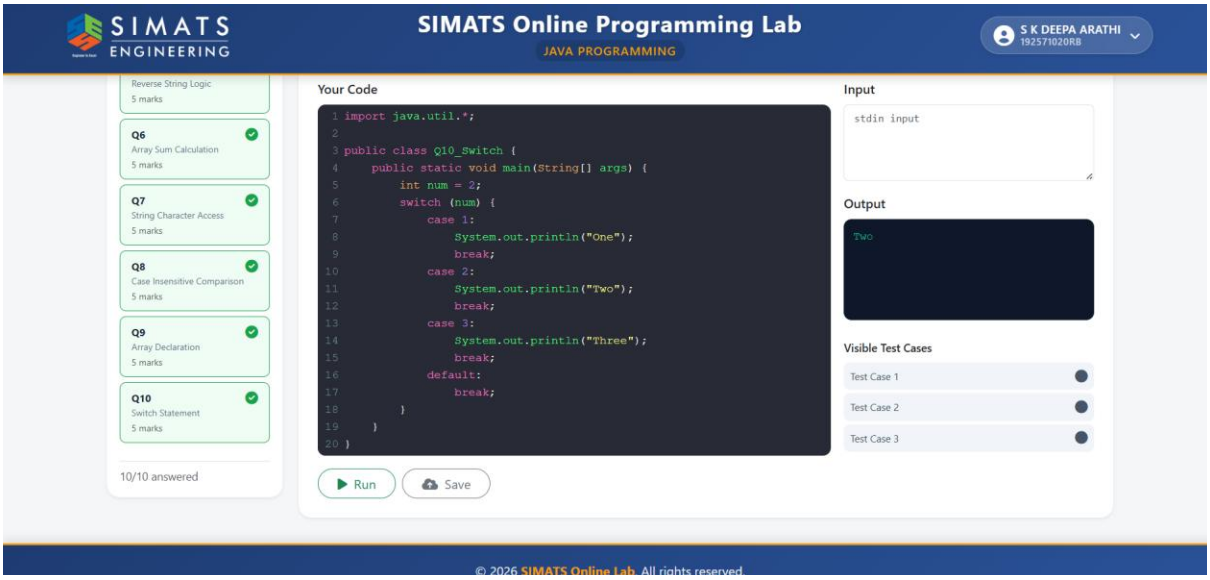
Q9. The program contains an error in array declaration and initialization. Identify the syntax issue and correct it.

```
import java.util.Arrays;

public class Q9_ArrayDeclaration {
   public static void main(String[] args) {
      int arr[] = {1,2,3};
      System.out.println(Arrays.toString(arr));
   }
}
```



Q10. A switch statement prints multiple outputs due to missing statements. Identify the problem and correct the code.

```
public class Q10_Switch {
    public static void main(String[] args) {
        int num = 2;
        switch(num){
            case 1: System.out.println("One"); break;
            case 2: System.out.println("Two"); break;
            case 3: System.out.println("Three"); break;
        }
    }
}
```



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach

Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases